

MORINGA SCHOOL

AI ESSENTIALS AND WORKFLOW AUTOMATION COURSE

AUTOMATED FOREX INTELLIGENCE & PRICING PLATFORM

WEEK 10 CAPSTONE PROJECT

README DOCUMENTATION

SUBMISSION DATE: 8TH JUNE 2026

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1. Project Structure

Main Financial Insights Workflow

Responsible for the end-to-end processing of financial intelligence data including:

- Scheduled workflow execution
- Retrieval of financial and forex data from multiple sources
- Source-specific formatting and extraction
- AI-powered summarization using Mistral AI
- Structured output validation using JSON parsers
- Validation checks before merging data
- Consolidation of validated source outputs
- Financial comparison dashboard preparation
- Final newsletter formatting
- Automated email distribution through Gmail
- Newsletter archiving into Google Sheets
- Fallback AI handling where required

Error Handling Workflow

Responsible for workflow-level error monitoring and operational alerting including:

- Detection of workflow execution failures using an Error Trigger node
- Logging workflow errors into Google Sheets
- Sending automated Gmail error notifications
- Supporting operational visibility and troubleshooting

Documentation Components

The project documentation package includes:

- README Documentation
- Formal Project Documentation
- Presentation Slides
- Workflow screenshots
- Execution evidence screenshots

Workflow Export Files

The project repository includes the following exported workflow files:

- Main Financial Insights Workflow JSON
- Error Handling Workflow JSON

2. Installation Instructions

Step 1: Download Project Files

Download all workflow JSON files and documentation files from the GitHub repository.

Step 2: Import Workflows into n8n

Import the following workflows into the n8n environment:

- Main Financial Insights Workflow
- Error Handling Workflow

Step 3: Configure Credentials

Configure the following credentials inside n8n:

- Mistral AI Credentials
- Gmail Credentials
- Google Sheets Credentials

Step 4: Configure Financial Data Sources

Update all HTTP Request nodes with the required financial source URLs and authentication details where applicable.

Configured financial sources include:

- FloatRates
- NCBA
- Nairobi Securities Exchange (NSE)
- Central Bank of Kenya (CBK)
- Bank of Africa
- ExchangeRate API

Step 5: Activate Workflows

After validation and testing, activate all workflows within n8n.

3. Setup & Configuration Steps

Financial Source Configuration

Configure all HTTP Request nodes to retrieve financial intelligence data from the required financial platforms.

The workflow retrieves and processes financial information including:

- Forex exchange rates
- Financial market updates
- Banking and market intelligence
- Currency movement summaries
- Market outlook insights

AI Model Configuration

Configure the Mistral AI model within all AI Agent nodes.

The workflow uses:

- Mistral AI for financial summarization
- Structured Output Parser nodes for JSON validation
- Validation IF nodes for structured output reliability

Logging Configuration

Configure Google Sheets logging nodes to capture operational execution data including:

- Workflow execution timestamps
- Source success/failure status
- Number of processed financial sources
- AI execution status
- Workflow execution outcomes
- Error details

Email Configuration

Configure Gmail integration and update recipient email addresses for automated financial insights distribution.

The workflow generates automated email briefings containing:

- Forex dashboard tables
- Financial summaries
- Market outlook insights
- Operational execution outputs

Error Handling Configuration

Configure the Error Handling Workflow to support operational monitoring and recovery visibility.

The workflow should be configured to:

- Automatically trigger during workflow execution failures
- Capture and log workflow error details into Google Sheets
- Send automated Gmail notifications when failures occur

Validate all Google Sheets and Gmail credentials before activating the workflow.

GitHub Repository Configuration

Upload the following project assets to the GitHub repository:

- Workflow JSON exports
- README Documentation
- Formal Project Documentation
- Presentation Slides
- Workflow screenshots
- Execution evidence screenshots

4. Environment Requirements

Software Requirements

- n8n instance (Cloud or Self-Hosted)
- Modern web browser
- Internet connectivity

API & Service Requirements

- Mistral AI API Key
- Gmail integration credentials
- Google Sheets integration credentials
- Access to financial data source websites

5. Usage Guide

Workflow Execution

The workflow executes automatically using scheduled triggers configured within n8n.

During execution, the system performs the following actions:

1. Retrieves financial intelligence data from configured financial websites
2. Validates source availability and response quality
3. Processes and structures retrieved financial data
4. Generates AI-powered financial summaries using Mistral AI
5. Validates structured AI outputs
6. Merges validated outputs into a unified financial briefing
7. Logs operational execution details into Google Sheets
8. Sends automated financial insights emails to recipients

Financial Insights Email Output

The automated financial email briefing includes:

- Forex dashboard table featuring currency exchange rate summaries
- AI-generated financial market summaries
- AI-generated market outlook insights

Error Handling Workflow Usage

The Error Handling Workflow is configured as a separate operational recovery workflow within n8n.

When a workflow-level failure occurs, the Error Trigger node automatically activates the recovery sequence.

The Error Handling Workflow performs the following actions:

1. Detects workflow execution failures
2. Captures operational error details
3. Logs error information into Google Sheets
4. Sends automated error notification emails through Gmail

This implementation improves operational visibility and ensures workflow failures are recorded and communicated without requiring manual monitoring.

Operational Logging Usage

The workflow logs operational execution data into Google Sheets including:

- Execution timestamps
- Source success/failure status
- AI execution status
- Workflow execution outcomes
- Error details
- Operational recovery actions

These logs support operational auditing, troubleshooting, and governance visibility.

6. Deployment Instructions

Deployment Process

1. Import all workflow JSON files into n8n
2. Configure all required credentials
3. Configure financial source endpoints
4. Validate all workflow execution paths
5. Test structured output validation
6. Test error handling execution
7. Activate production workflows

Recommended Deployment Sequence

Deploy and validate workflows in the following order:

1. Main Financial Insights Workflow
2. Error Handling Workflow

After deployment:

- Test workflow execution success scenarios
- Simulate workflow failures
- Validate Google Sheets error logging
- Validate Gmail error notification delivery

7. Troubleshooting Section

Issue	Possible Cause	Resolution
Financial source unavailable	Website/API outage	Workflow continues using available sources
Invalid AI response	Malformed structured output	Review Structured Output Parser Configuration
Email delivery failure	Gmail authentication issue	Reconfigure Gmail credentials
Missing operational logs	Google Sheets connection issue	Reconnect Google Sheets integration
Workflow execution failure	Workflow-level processing issue	Review Error Handling Workflow logs and Gmail notifications
Error notifications not received	Gmail node configuration issue	Validate Gmail credentials and recipient configuration